10E — JianghuSect 2 (JS2-v2.6.0) Deep-Dive Feature Audit: Economy, Supply Chains, CP & External Trade (Part 5 of 5 — Grand Finale: Master Closed-Loop Synthesis, 40-Script Wallet Conservation, EC-08 Peak Pill Conjuration & P0 $\to$ P3 Roadmap)

Target Codebase: JS2-v2.6.0 (Holistic synthesis across economy.js, cpeco.js, external_market.js, sect.js, sim.js, ai.js, cultivation.js, expedition.js, explore.js, content.js) **Target Audience**: Lead AI Systems Architects & Autonomous Coding Agents (**GPT** & **Gemini**) **Immutable Project Rules**:

1. **NEVER modify `js/human_core.js`** (HC.FINGERPRINT === '155:2f9c9dd7').
2. **Preserve exact 40 `<script>` tags** and <= 3.5 ms/tick 80-NPC performance budget.
3. **Global Closed-Loop Invariant**: Across every tick, day, month, and year of natural simulation (excluding explicit Player Edit-Hub spawns), total world Spirit Stones must obey strict conservation (Debit === Credit) across well-defined wallets!

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1. [ECO-SYS-01] Complete Multi-Wallet Map & All Remaining Faucets/Sinks Across the 40 Scripts

Currently, Spirit Stones are stored in **8 fragmented schemas** (2 of which are duplicate/split bugs!) and mutated ad-hoc across 9 files:

1.1 Canonical 6-Wallet Consolidation (Eliminating the 2 Split-Wallet Bugs)

1. **`worldPool`**: W.world.resourcePool.stones (Continental Spirit-Vein & Regional Market Pool)
2. **`sect`**: W.sect.stones (Central Sect Treasury)
3. **`peak:<peakId>`**: Unify pObj.treasuryStones and pObj.vault.stones into a single canonical pObj.treasuryStones (with getter/setter alias on vault.stones, fixing 07B)
4. **`npc:<npcId>`**: Unify npc.inv.stones and orphaned npc.personalStones (fixing 07A) into npc.inv.stones

5. **`visitor:<visId>`**: `visitor.stones` (Visiting Merchant working float, drawn from and returned to `worldPool` on arrival/departure, `10D`)
6. **`rival:<sectId>`**: `rivalSect.treasury` (`19+` Rival Sects, funded seasonally from `worldPool` by node ownership/tier, `08A`)

1.2 Master Audit Table of Unclosed Faucets & Black-Hole Sinks Still Live in `JS2-v2.6.0`

Direction	Location (`File : Function`)	Current Broken Behavior	Exact Closed-Loop Fix
Faucet (+Δ)	`economy.js` : `ECO.mineYield` & `ai.js:592` `A.mine.done`	Mints infinite stones + `(0 || 1)*6` personal stones with `0` pool debit (`10A`)	Debit `worldPool` (with bounded seasonal Leyline/Celestial vein recharge)
Faucet (+Δ)	`sect.js` : `quartermaster` & `auditOfficerConduct`	Mints Day-0 logistics stones & `+45` confiscation from `0`-stone offenders (`07A`)	Require real tributary/convoy source; cap confiscation by `offender.inv.stones`
Faucet (+Δ)	`explore.js` : `dispatchExploration` (`OUT-05`) & `solveElementalPuzzleGate`	Credits `+820` / `+400` stones even when `worldPool === 0` (`08C`)	Clamp `actual = Math.min(raw, worldPool)` (`debit === credit`)
Faucet (+Δ)	`expedition.js` : `PILLAGE` & `collectVassalTributes` (`OUT-19`)	Formula-mints stones exceeding `rivalSect.treasury`; repeats on `day === 0` (`08A`)	Bound by `rivalSect.treasury`; guard seasonal `periodId`
Faucet (+Δ)	`external_market.js:186` : `EXT.openVisit`	Conjures `280–3,000` `v.stones` out of thin air (`10D`)	Draw float from `worldPool` on arrival
Sink (−Δ)	`external_market.js:505` : `EXT.daily` departure	Deletes departing `v.stones` (destroying all stones	Return remaining `v.stones` to `worldPool` on departure

Direction	Location (`File : Function`)	Current Broken Behavior	Exact Closed-Loop Fix
		paid by our sect!) (`08F/10D`)	
Sink (`-Δ`)	`ai.js:893 AI.kill` & `sim.js:347 SIM.retire`	Destroys `50%` (death) or `100%` (retire/exile) of `n.inv.stones` (`07E/08B`)	Transfer `100%` (`heirs/sect` on death; `personal bag` to rival/wanderer or confiscated on exile)
Sink (`-Δ`)	`sect.js` : `payOfficeStipends`	Drains `~680/mo` from `W.sect.stones` into dead `officer.personalStones` (`07A`)	Credit `officer.inv.stones`
Sink (`-Δ`)	`sect.js` / `sim.js` / `cultivation.js` / `expedition.js`	Building upgrades, monthly upkeep, panic grain (`sim.js:518`), `formDemonPact` (`160`), `anchorDragonVeinNode` (`120`), `brewMultiphaseElementalPill` (`20`), caravan manual (`680`), `beginCelestialTribulation` (`500`) vanish	Credit `worldPool` (regional labor/materials/energy recirculation!)

1.3 Central Atomic Transfer Kernel (`ECO.transferStones`) & `v2.6.0` Digest Reconciliation

Implement `ECO.transferStones(W, { from, to, amount, why, allowPartial, txKind })`:

- Atomically verifies available balance in `from` wallet, clamps if `allowPartial`, deducts `actual` from `from`, and credits `actual` to `to`.

- Automatically classifies `W.sect.ledger` entries whenever `from === 'sect'` or `to === 'sect'`:
  - **External Cashflow (`internalTransfer: false`)**: `sect <-> worldPool | visitor:* | rival:*` (counted in `computePeriodDigest` `stoneIncome` / `stoneExpense`!).
  - **Internal Transfers (`internalTransfer: true`)**: `sect <-> peak:* | npc:*` (`salaries, stipends, stone_grant, death inheritance, embezzlement recovery` — tracked cleanly in `computePeriodDigest` `internalStoneNet` so **`openingStones + netExternalStones + netInternalStones + playerSpawnedStones === closingStones` holds to the exact integer 100% of the time!**).

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2. `[ECO-SYS-02]` Unfixed `EC-08` Peak Double-Spend Pill Conjuration (`js/sect.js:11935–12056`)

- **Exact Bug Verified in `EC-08`**:
  - Calling `SECT.PEAKS.allocateCentralResourcesToPeaks(W, { poolStones: 350 })` transfers `350` Spirit Stones into Peak treasuries **AND simultaneously adds `pillReserve += Math.floor(350 / 35)` (`+10` pills)**, which then distributes real `P:BASIC_QI` / `P:BASIC_HP` pills into disciples' bags **even when `W.sect.stock` has `0` pills and `0` herbs were consumed**! One Spirit Stone is spent twice—once as cash and once as conjured medicine!
- **Exact Fix**:
  - Separate stone budget transfer (`ECO.transferStones(W, { from: 'sect', to: 'peak:' + peakId, ... })`) from pill allocation: Peak `pillReserve` must **only** draw real existing pills above `ECO.reserveFloor` from `W.sect.stock` (`ECO.take(W, pillKey, qty)`), never conjuring pills from a division formula!

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3. `[ECO-SYS-03]` Cross-Currency (`CP <-> Spirit Stones <-> Goods`) No-Arbitrage Invariant

Once `stone_grant` (`10C`) is anchored to `c.price * c.markup` (`grantPrice = ceil((stones / 1.8) * c.price * c.markup)`) and gated behind `tg.skillOk === true`:

1. **Loop A (`Sect Shop -> Donate`)**: `CPE.priceOf(id) >= ceil(CPE.reward(id) * 1.3) + 2` $\implies$ max return `<= 0.76×` (`strictly dissipative`).

2. **Loop B (`CP Inputs -> Personal Craft -> Donate`)**: `donatePersonalProduct` caps reward at `0.88 * storeBoughtCost` (`strictly dissipative`).
3. **Loop C (`CP -> stone_grant -> Merchant Buy -> Donate for CP`)**:
    - Spend `100 CP` at `c.price = 1.0` $\to$ `<= 180` earmarked track stones (locked to `checkTrackGates` consumption; even if compared at theoretical value: `180` stones buys `180 / (base * 1.15)` items $\to$ donates for `base * 0.22 * sc` CP $\le 34..86`CP`< 100`CP across all`c.price ∈ [0.6, 4.0]`)!

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4. Master Prioritized Execution Roadmap for GPT & Gemini (`P0` $\to$ `P3` Across `10A–10E`)

Phase	Priority	Target Files	Exact Deliverables Across `10A–10E`
Phase 1	**P0 — Instant Gear Deletion, Aborted-Craft Material Loss & `stone_grant` Drain**	`economy.js`, `external_market.js`, `cpeco.js`, `ai.js`	1) Stamp `listedAtDay = W.day` in `ECO.mintGear` and guard `boughtKeys` in `EXT.sellStaleSectGear` to stop 0-ms buy-then-dump deletion (`10D`). 2) Refund reserved `craftTransactions` inputs when `AI.cancelAct` interrupts an in-progress `brew`/`forge`, and reserve `personalBrew` inputs upfront at `start` (`10B/10C`). 3) Fix `stone_grant` to respect `cashReserve`,

Phase	Priority	Target Files	Exact Deliverables Across `10A–10E`
			require `tg.skillOk === true`, scale with `c.price * c.markup`, and log internal transfers (`10C`).
Phase 2	**P1 — Atomic `ECO.transferStones` Kernel, `EC-06/08/09/10` & Manual Confiscation**	`economy.js`, `sect.js`, `external_market.js`, `cpeco.js`, `sim.js`	1) Implement `ECO.transferStones` unifying the 6 canonical wallets (`merging personalStones -> inv.stones & vault.stones -> treasuryStones`), recycling `visitor.stones` $\leftrightarrow$ `worldPool`, and reconciling `v2.6.0 computePeriodDigest` (`10E`). 2) Fix `EC-06` (propagate `playerSpawned` taint to crafted outputs), `EC-08` (Peak `pillReserve` draws real stock only), `EC-09` (`surplusRecipes` checks `v.wants`), and `EC-10` (`retireGear` single-salvage guard) (`10B/10D/10E`).

Phase	Priority	Target Files	Exact Deliverables Across 10A–10E
			3) Grant studyWaivers instead of confiscating member-bought/duplicate personal manuals at midnight, and filter dur <= 0 broken gear out of CPE.shopItems/pickPurchase (10C/10D).
Phase 3	**P2 — 3-Queue Workshops, 475 Expansion Recipes, Bargaining & Stagflation Fix**	economy.js, cpeco.js, external_market.js, sect.js	1) Split orders.forge into orders.forge (FORGING) + orders.array_talisman (ARRAY/TALISMAN), sync ECO.recipesFor with all 1,370 recipeById recipes, and remove high-tier recipe penalty when elders need upgrades (10B). 2) Wire real memberTradeSkill + marketmaster + FREE_TRADE_TREATY into EXT.quotePrice, blend 0.45*avg + 0.55*median in CPE.daily, pay 30%

Phase	Priority	Target Files	Exact Deliverables Across 10A–10E
			teacher honorarium in guidedStudy, and bridge organs into donorValue/A.donate (10A/10C/10D).
Phase 4	**P3 — Xianxia Bigu Fasting, A.personalForge, Craft XP & Elder Lineage Sinks**	ai.js, economy.js, cpeco.js	1) Scale high-stage Bigu (辟谷) satiety decay (0.20× at Stage 7–9), protect mealSpecialties from dry-ration bulk takes, and raise Autumn grain reserves (10A). 2) Add A.personalForge for FORGING/TALISMAN/ARRAY, scale grantCraftCultivation with hours worked, allow Lv.4–5 mines/gardens rare Tier-4 procs, and add Elder Mentor-Sponsorship CP sinks (10A–10C).